How the agents are wired, and what each one remembers

Five figures. Read them in order: where each LangGraph pattern lives, how a request moves through the two busiest workflows, where each memory layer sits, and how learning is allowed to happen.

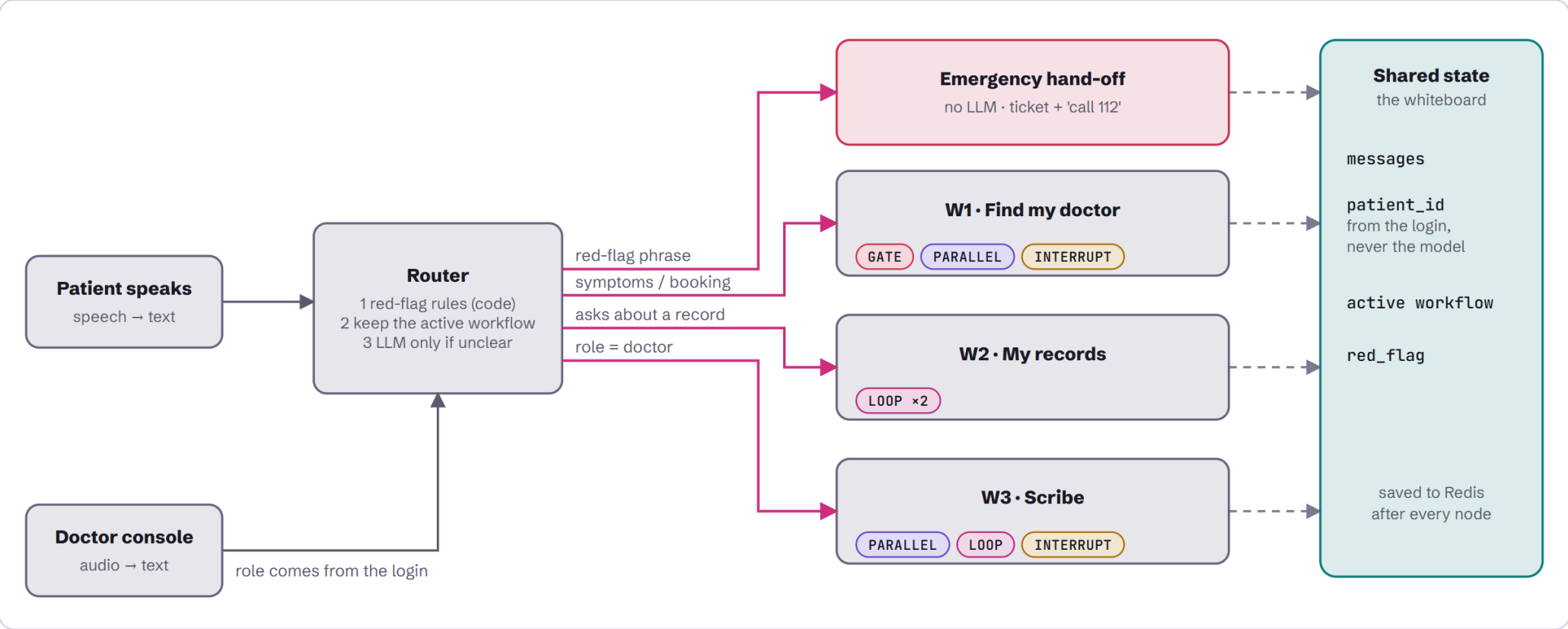
plain code model call tool call waits for a human storage safety gate

1. A router first, then four paths
2. Find my doctor, step by step
3. Records and scribe verify themselves
4. Where each kind of memory lives
5. How the memory teaches itself

Generated from the Sahyog Care Companion design. LangGraph routing, workflow patterns and memory strategy, with Redis for thread, patient and agent memory and Postgres as the system of record.

1. A router first, then four paths

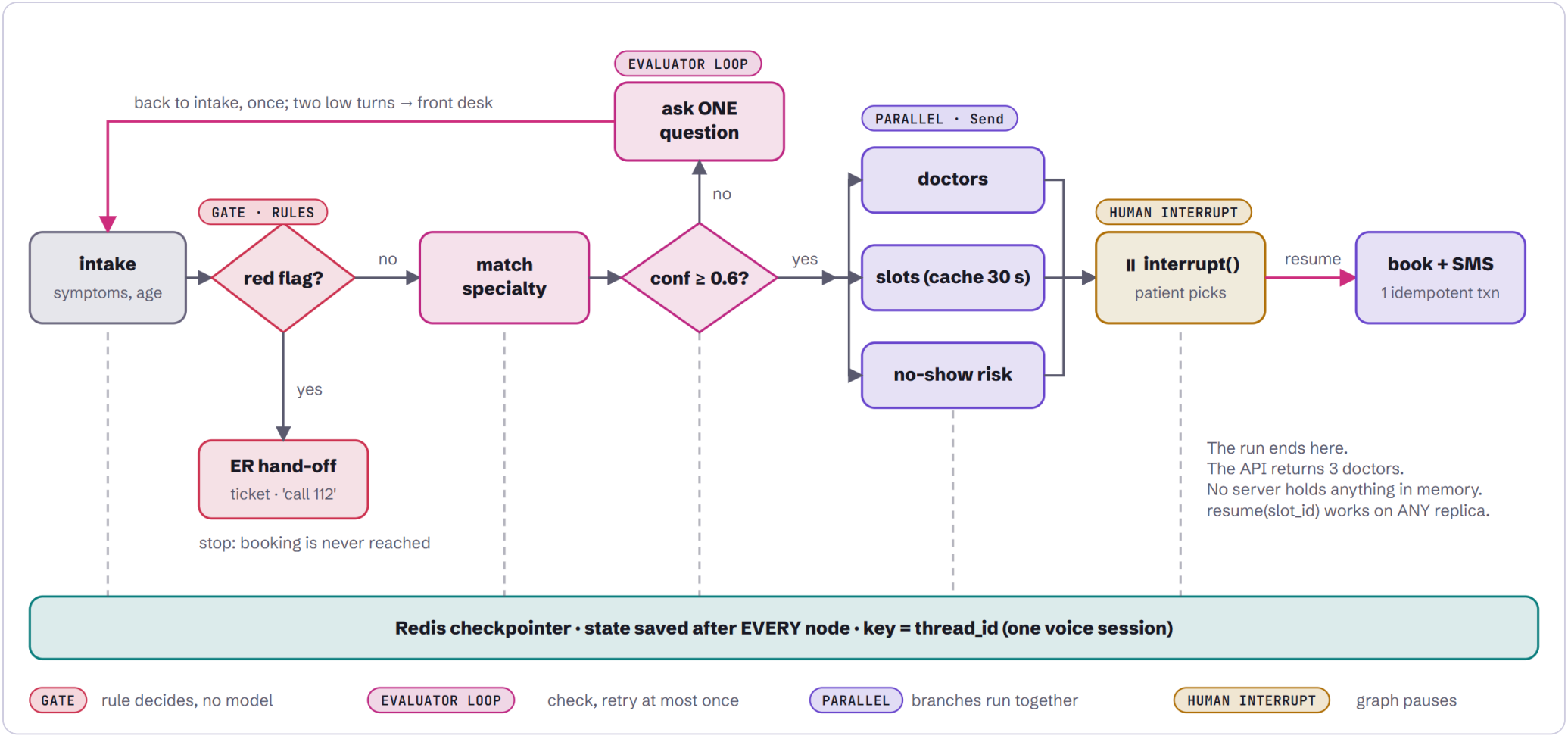
Which pattern runs at the top of the system?



Routing sits at the top. Rules decide the common cases for free, a remembered *active workflow* keeps the conversation with the same agent, and an LLM is asked only when the intent is truly unclear. Every path reads and writes the same shared state.

2. Find my doctor, step by step

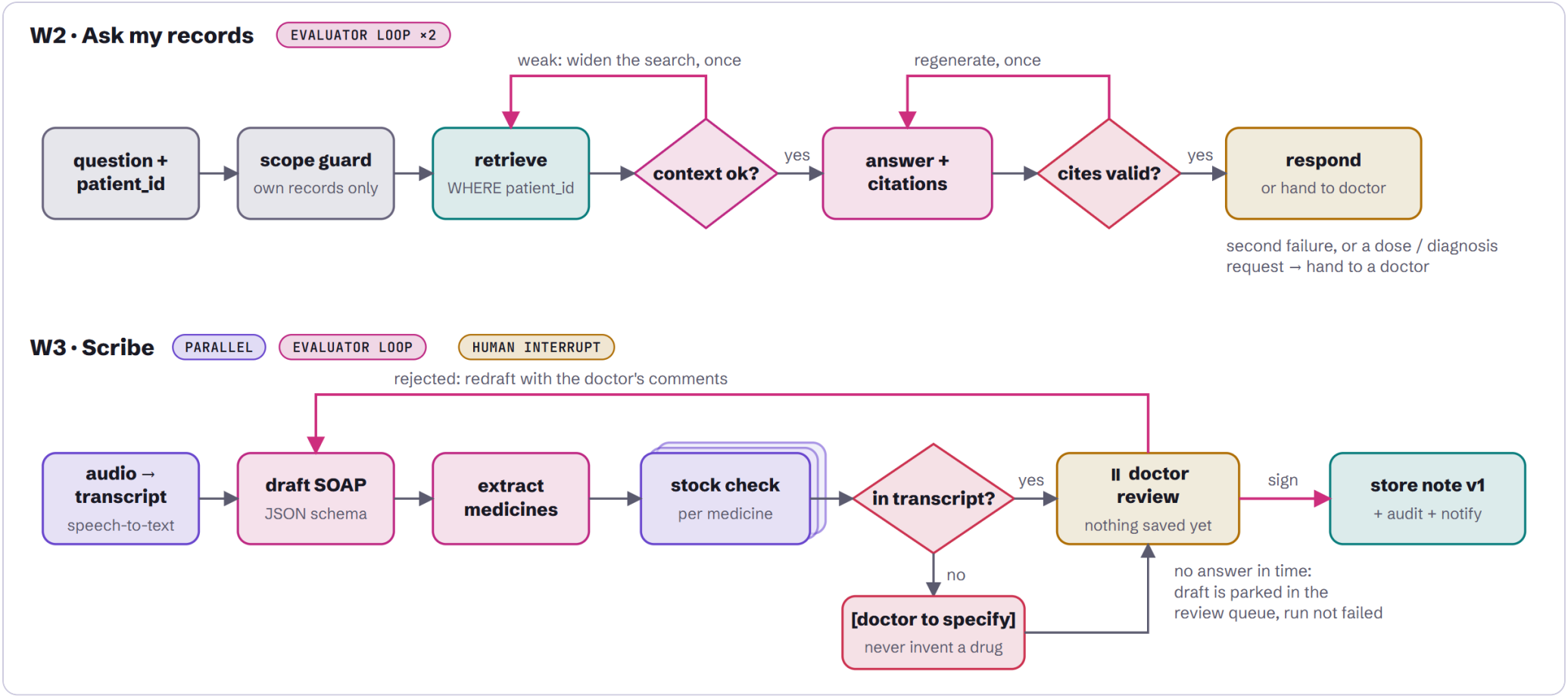
Where do the gate, the loop, the parallel step and the pause go?



One chain with four tools. The red-flag gate is code. The clarifying question is a bounded loop. Doctors, slots and risk are fetched at the same time. The pause saves everything to Redis, so the API call returns and any server can resume.

3. Records and scribe verify themselves

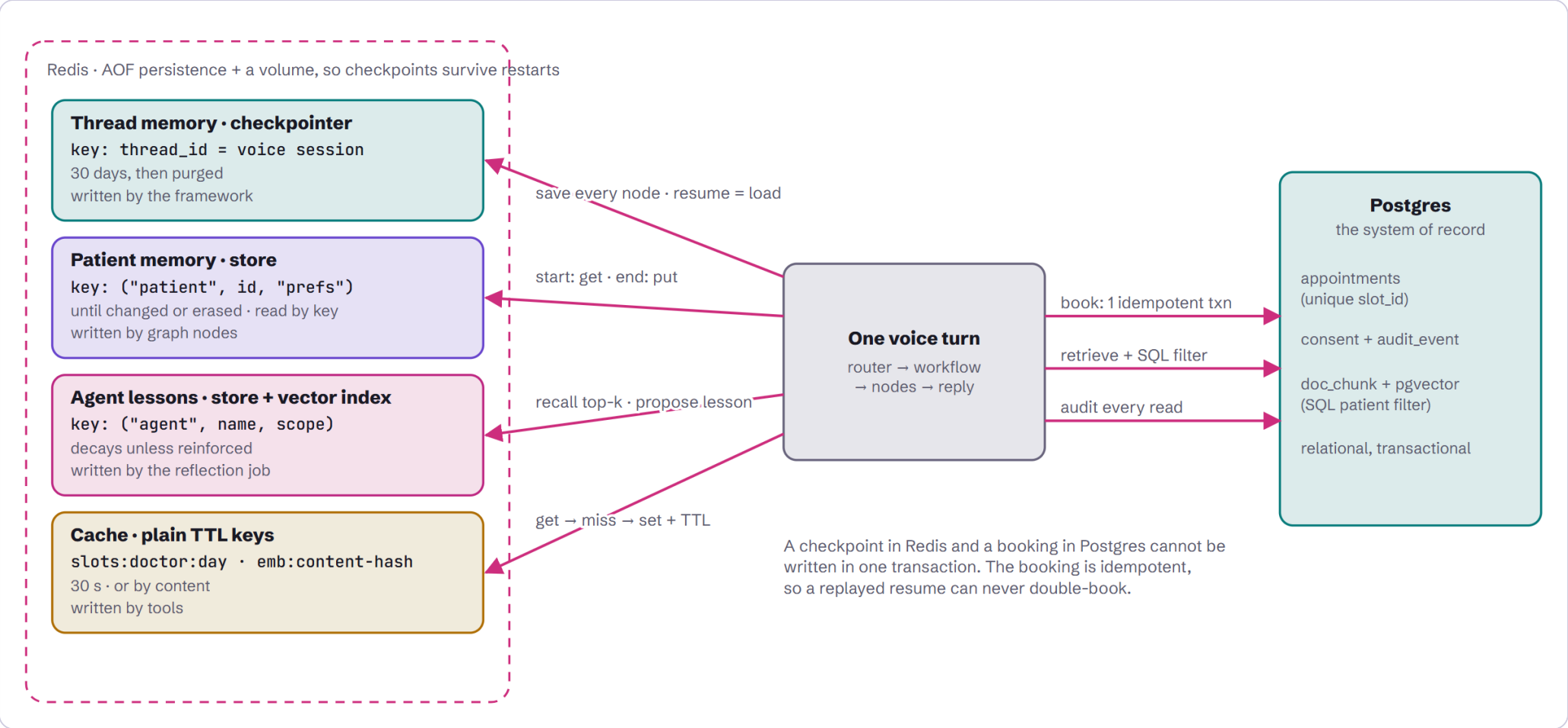
Where does the evaluator-optimizer pattern earn its latency?



Each loop runs once, never twice. The records flow retries a weak search and a bad citation. The scribe never retries a drug: if a medicine is not in the transcript it becomes *[doctor to specify]*, and a person signs before anything is saved.

4. Where each kind of memory lives

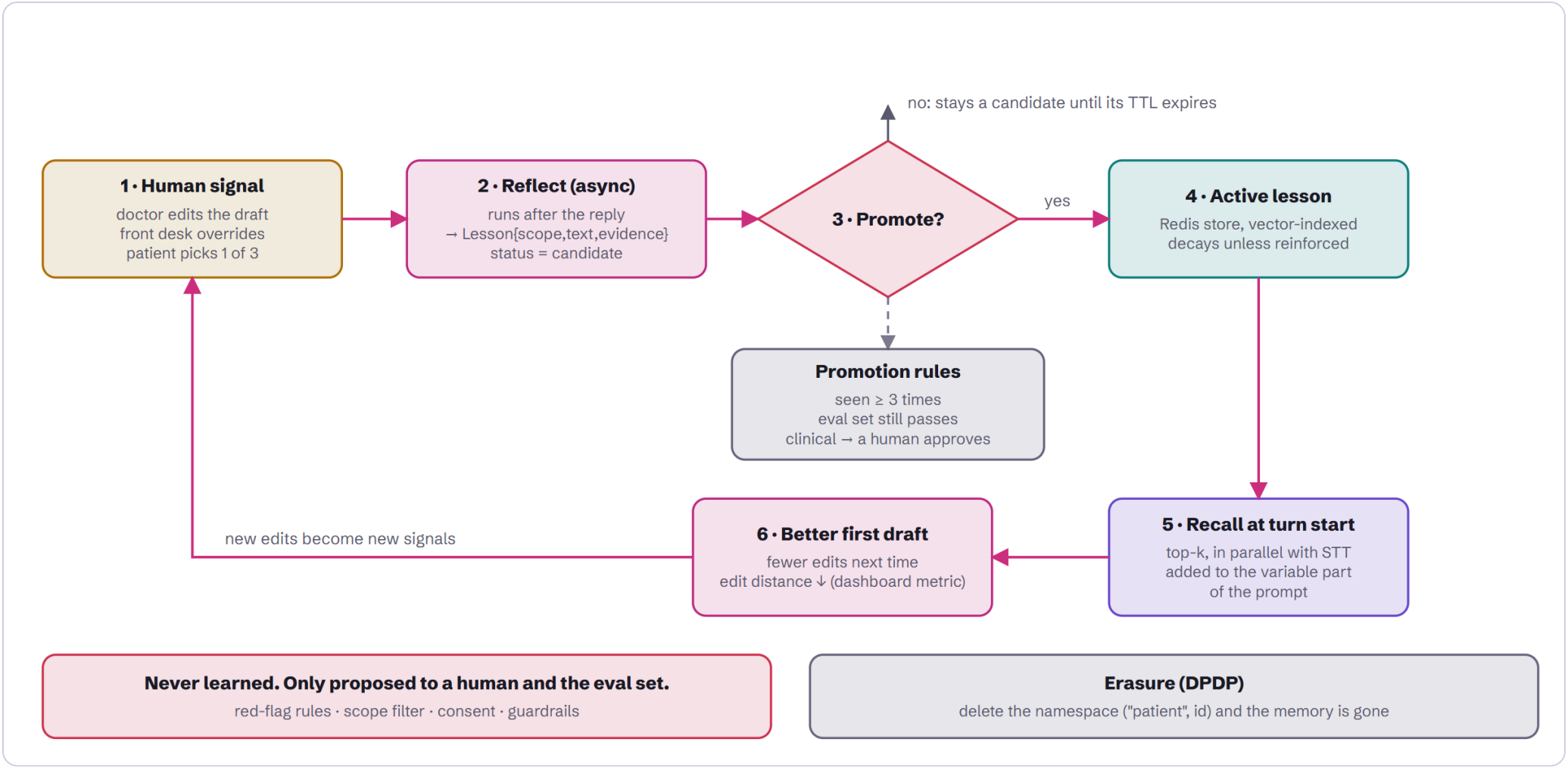
Which memory is Redis, which is Postgres, and when is each touched?



Four layers in Redis, one system of record in Postgres. Thread memory is the checkpointer and also what people call short-term memory. Patient and agent memory are the long-term store, split by namespace. The cache is just keys with a lifetime.

5. How the memory teaches itself

What stops a learning system from learning something dangerous?



Humans and tests sit between a signal and an active lesson. Edits and overrides become candidates, a repeat count plus the eval set plus a human for clinical items promotes them, and the safety rules are outside the loop entirely. The learning shows up as falling edit distance.